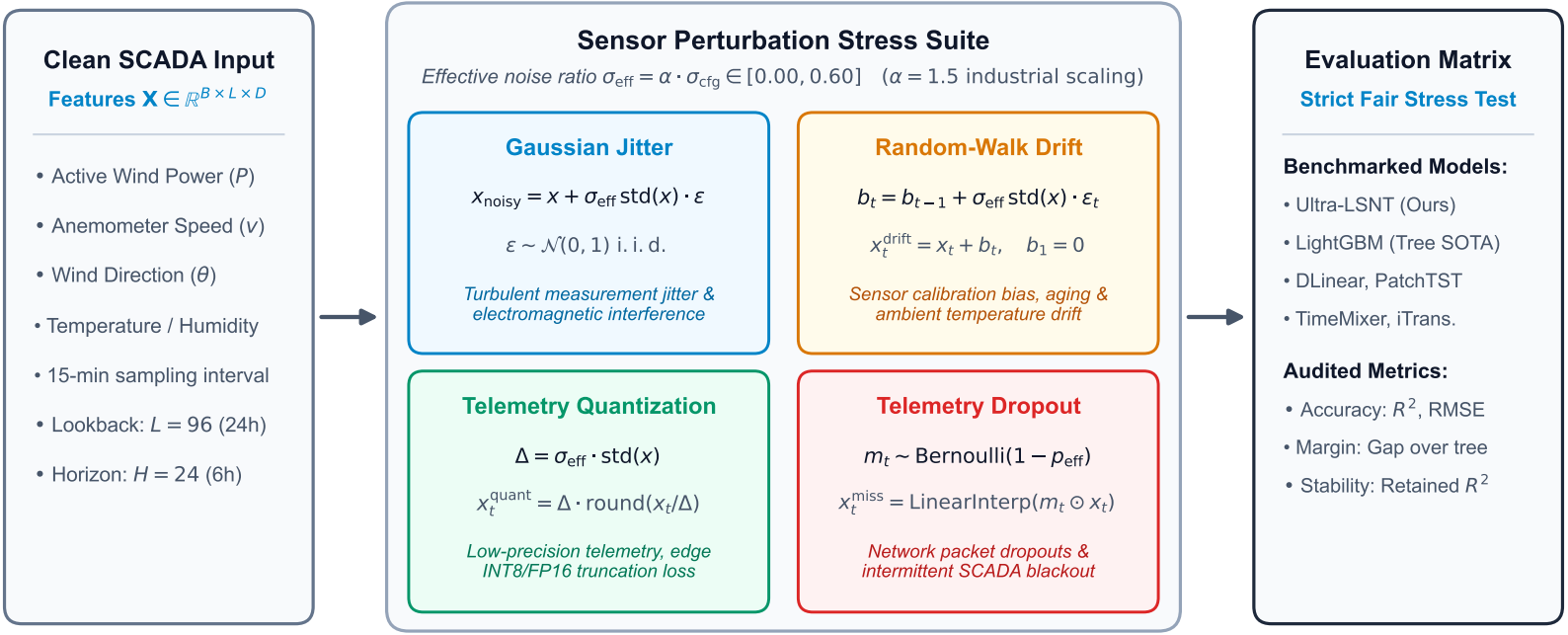

Clean SCADA Input
Features $\mathbf{X} \in \mathbb{R}^{B \times L \times D}$
• Active Wind Power (P)
• Anemometer Speed (v)
• Wind Direction (θ)
• Temperature / Humidity
• 15-min sampling interval
• Lookback: $L = 96$ (24h)
• Horizon: $H = 24$ (6h)
Sensor Perturbation Stress Suite
Effective noise ratio $\sigma_{\text{eff}} = \alpha \cdot \sigma_{\text{cfg}} \in [0.00, 0.60]$ ($\alpha = 1.5$ industrial scaling)
Gaussian Jitter
$x_{\text{noisy}} = x + \sigma_{\text{eff}}\,\text{std}(x) \cdot \varepsilon$
$\varepsilon \sim \mathcal{N}(0, 1)$ i. i. d.
Turbulent measurement jitter & electromagnetic interference
Random-Walk Drift
$b_t = b_{t-1} + \sigma_{\text{eff}}\,\text{std}(x) \cdot \varepsilon_t$
$x_t^{\text{drift}} = x_t + b_t, \quad b_1 = 0$
Sensor calibration bias, aging & ambient temperature drift
Telemetry Quantization
$\Delta = \sigma_{\text{eff}} \cdot \text{std}(x)$
$x_t^{\text{quant}} = \Delta \cdot \text{round}(x_t/\Delta)$
Low-precision telemetry, edge INT8/FP16 truncation loss
Telemetry Dropout
$m_t \sim \text{Bernoulli}(1 - p_{\text{eff}})$
$x_t^{\text{miss}} = \text{LinearInterp}(m_t \odot x_t)$
Network packet dropouts & intermittent SCADA blackout
Evaluation Matrix
Strict Fair Stress Test
Benchmarked Models:
• Ultra-LSNT (Ours)
• LightGBM (Tree SOTA)
• DLinear, PatchTST
• TimeMixer, iTrans.
Audited Metrics:
• Accuracy: R^2, RMSE
• Margin: Gap over tree
• Stability: Retained R^2